

OpenVAF — Verilog-A LRM Compliance

Clause-by-clause coverage of the Verilog-A language (VAMS LRM 2023, Annex C)

Ngspice + OpenVAF Enhancements project

July 2026

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OpenVAF — Verilog-A LRM Compliance

How the `openvaf-r` compiler in this repository covers the Verilog-A language, clause by clause against the Verilog-AMS Language Reference Manual (the 2023 edition, [docs/VAMS-LRM-2023.pdf](#); Verilog-A is its Annex C analog-only subset). Every claim in this document is backed by a committed, executable check: the code examples are drawn from the verify suites under [examples/](#), which compile each construct with the committed compiler, simulate it with the committed ngspice, and compare the numbers against closed-form expectations on every regression run.

Status marks used throughout:

Mark	Meaning
✓	fully implemented and runtime-verified
⚠	implemented with LRM-sanctioned fallback semantics, or a documented deviation
×	out of scope: a Verilog-AMS (mixed-signal) feature outside Annex C

A one-page summary table closes the document. Companion references: the [User Handbook](#) (task-oriented), the [openvaf change report](#) (where each capability was implemented and why), and the per-enhancement write-ups in [enhancements_doc/](#).

1. Compilation model and scope (LRM 1, Annex C)

OpenVAF compiles one Verilog-A source file (with its ``include` closure) into an OSDI shared object: analytic residuals and Jacobians are generated by automatic differentiation, so every construct below is “supported” only if it produces correct derivatives as well as correct values — a stricter bar than parsing. Hierarchy is resolved by a compile-time elaboration pass (§7), after which the pipeline sees a single flat module.

Annex C draws the language boundary this project targets: the analog subset. Digital and mixed-signal constructs (reg/wire logic, digital `always/initial`, connect modules, ``default_discipline`, discrete-domain disciplines) are × out of scope by definition, and the compiler rejects them with located diagnostics rather than accepting them silently — e.g. `domain discrete` on a discipline with natures is a proper two-label error citing LRM 3.6.2.2.

2. Lexical conventions (LRM 2) — ✓

Identifiers, including escaped identifiers (LRM 2.7.1):

```
electrical \node-1a ;           // any printable characters until whitespace
parameter real \my.gain = 2.0;
```

Numbers (LRM 2.5): decimal integers, reals with exponents, SI scale suffixes (1k, 2.5u, 10n), and based integer literals in all four bases with optional size and sign, underscores as separators:

```
integer a, b, c, d;
analog begin
  a = 8'hFF;           // sized hex           -> 255
  b = 'sb1010_1100;    // signed binary with separators
  c = 'o17;            // octal               -> 15
  d = 4'd12;           // sized decimal
end
```

The don't-care digits x/z/? are additionally accepted in the power-of-two bases when the literal is a casex/casez item (§5.4); anywhere else they are a located error rather than silent zeros.

Strings (LRM 2.6.3) with the full escape set — \n, \t, \\, \", and octal \ddd — processed by a single-pass unescaper (a sequential-replacement implementation corrupted overlapping escapes and was rewritten). Comments both styles, including the edge case of a // comment at end-of-file without a trailing newline (formerly a lexer hang). Attributes (* ... *) parse everywhere the LRM allows and carry simulator-facing metadata (§9.6).

3. Data types (LRM 3)

3.1 Value types — ✓

integer (32-bit), real (IEEE double), and string variables, with genuine persistence across evaluations for both numeric types (a Verilog-A module variable keeps its value between solver calls, the foundation of event counters and iterative models):

```
integer count;           // persists across evaluations
analog begin
    @(cross(V(in) - 0.5, +1))
        count = count + 1; // genuinely accumulates
end
```

Uninitialized strings default to ""; declaration initializers work on scalars and (multi-dimensional) arrays, evaluating once at setup when parameter-only:

```
real x = 2.0 * gain;
real taps[0:3] = '{0.1, 0.2, 0.3, 0.4};
```

3.2 Parameters (LRM 3.4) — ✓

Ranges and exclusions enforced on user-given values; localparam genuinely non-overridable; alias-param including \$param_given through the alias; string parameters (usable in case dispatch); array parameters of any dimensionality, expanded to per-element OSDI parameters that are individually overridable from SPICE:

```
parameter real r = 1k from (0:inf) exclude 1e6;
parameter integer sel = 0 from [0:15];
localparam real g = 1.0 / r;           // tracks r, not overridable
parameter real w[0:1][0:2] = '{1,2,3}, {4,5,6}';
// SPICE: .model m dev(w[1][2]=9.5)
```

One deliberate reading of the LRM, matching industry practice (the CMC convention): a parameter's default is exempt from its own range — compact models use out-of-range defaults as "feature disabled" markers — while every user-given value is fully validated. ⚠ documented in the [handbook's limitations chapter](#).

paramset blocks (a Verilog-AMS 3.4.6 feature adopted because real model libraries use them) work, including hidden-system-parameter assignments:

```
paramset fast_nmos nmos_va;
    .w = 2e-6; .l = 60e-9;
    .$mfactor = 4;
endparamset
```

3.3 Natures, disciplines, nets (LRM 3.5–3.7)

✓ Discipline/nature declarations with attribute access from expressions (`net.potential.abstol`); derived natures — nature `Xf` : Flow and deriving from a discipline's nature (: `electrical.flow`) — with full attribute inheritance; ground declarations in both orderings; vectored nets and ports with bit-selects; net initializers that become solver nodesets; domain `continuous`:

```
nature CurrentX : Current;    // derived nature, inherits abstol/units
    abstol = 1e-15;
endnature

electrical [3:0] bus;         // vectored net
electrical vout = 2.5;        // initializer -> OSDI nodeset (initial guess)
ground electrical gnd;
```

Signal-flow disciplines (potential-only or flow-only) are fully usable, including probe-only branches — probing a branch that is never contributed to reads its true flow (an ideal ammeter) instead of the 0-and-open-circuit a naive topology gives.

Named port branches (LRM 3.7.2) work as of E-84 and carry the same defining equation as a direct port-flow probe; contributing to one is a proper diagnostic (port branches are probe-only per the LRM):

```
branch (<p>) probe_p;         // named branch over port p
iprobe = I(probe_p);         // same value as I(<p>)
```

4. Expressions (LRM 4)

4.1 Operators and precedence (LRM 4.1, Table 4-2) — ✓

The full operator surface — arithmetic (incl. `**`), comparison, logical, bitwise, shifts (`<</>>` and arithmetic `<<</>>>`), ternary (including over strings), concatenation `{a,b}` and replication `{n{x}}` — was audited operator-by-operator and level-by-level against Table 4-2 with self-checking bitmask modules whose failure modes are numerically visible. Two real defects found and fixed in the audits: unary `~` emitted arithmetic negate, and the constant folder disagreed with the runtime on `>>`; one precedence defect: `%` bound tighter than `*/` (so `6*7%4` evaluated to 18 instead of 2). Associativity verified including `2**3**2 = 64` (left-associative per Verilog) and unary binding above `**` (`-2**2 = 4`).

4.2 Built-in and environment functions (LRM 4.3, 9.7) — ✓

`ln/log/exp/sqrt/pow/min/max/abs/floor/ceil`, trigonometric and hyperbolic families (integer `min/max/abs` correctly integer-typed; integer division/rounding away from zero per the LRM), and the environment: `$temperature` (kelvin — verified as the exact kT/q law across a $-50\dots150\text{ }^{\circ}\text{C}$ sweep), `$vt` and `$vt(T)`, `$abstime`, `$realtime` (identical in the continuous-time analog context), `$simparam` and `$simparam$str`, `$param_given`, `$port_connected`, `$mfactor` and the position hidden parameters.

4.3 Analog operators (LRM 4.5) — ✓ (one documented deviation)

Every analog operator produces exact Jacobian contributions via autodiff. The full argument surface (trailing tolerances, optional arguments) was audited at runtime — compiling proves nothing about whether a trailing argument is honored.

```
// time derivative & integrals
I(p,n) <+ ddt(c * V(p,n));
phi = idtmod(K * V(vco), 0.0, 1.0);    // phase wrapping (VCO idiom)
x    = idt(V(in), 0.5, rst);           // reset form holds at ic

// delays and filters
```

```

V(out) <+ absdelay(V(in), 1n);
V(out) <+ transition(sel ? 1.0 : 0.0, 0, 10n, 5n);
V(out) <+ slew(V(in), 1e6, -1e6);           // LRM negative max_neg_rate
V(out) <+ laplace_np(V(in), '{1.0}', '{-6.283e3, 0.0}'); // complex pairs
V(out) <+ zi_nd(V(in), '{0.5, 0.5}', '{1.0}', 1u);

```

```
// small-signal & events
```

```

gm = ddx(Id, V(g,s));           // symbolic derivative
tc = last_crossing(V(in) - 0.5, +1);
V(out) <+ ac_stim("ac", 1.0, 90.0); // module AS the AC stimulus

```

- ddt, idt (with initial conditions that survive into transient, and the assert/reset forms stable enough for relaxation oscillators), idtmod (integrates unbounded internally, wraps the returned value), ddx with respect to potentials *and* flows: ✓
- absdelay(+maxdelay), transition(3/4/5-argument, `default_transition defaults, DC well-posed), slew (honoring the LRM's *negative* max_neg_slew_rate — the sign defect that made it ignore its input was found in the audit): ✓
- laplace_nd/np/zd/zp via exact state-space realization and zi_nd/np/zd/zp via bilinear transform, both with complex pole/zero pairs per the LRM's (re, im) vector convention and parameter-dependent coefficients: ✓
- last_crossing with simulator-side waveform history: ✓
- noise sources (LRM 4.6): white_noise, flicker_noise, noise_table, noise_table_log (inline or file data); operating-point-dependent factors and ddt()-shaped noise are exact with no extra matrix unknowns; same-named sources sum coherently per LRM 4.6.4 including anti-phase cancellation: ✓

```

I(a,b) <+ white_noise(4.0 * `P_K * $temperature / r, "thermal");
I(d,s) <+ gm * white_noise(gamma_4kT, "induced"); // op-dependent factor

```

- analysis("ac","tran",...) variadic, with the AC/noise operating point counting as its parent analysis per LRM 4.6.1: ✓
- \$limit (built-in "pnjlim" and user functions — genuinely engages SPICE-style iteration limiting), \$bound_step, \$discontinuity(n) (clamps the next step *and* makes the integrator bisect onto the event): ✓
- \$table_model (LRM 9.19 by clause, listed here as a modeling operator): 1-D piecewise-linear, N-dimensional multilinear, and natural cubic-spline interpolation — lowered to differentiable MIR so *all* partial derivatives feed the Jacobian (a table-based MOSFET gets exact gm and gds):

```
I(d, s) <+ $table_model(V(g, s), V(d, s), "mos_iv.tbl", "1l");
```

- $\triangle$ **limexp** is implemented stateless (exact exp below the overflow threshold, tangent-continued above): a stateful previous-iterate limiting version was built and *reverted* because correct converged values under limiting require the simulator's limiting-RHS correction, which applies to circuit unknowns, not to limexp's derived argument. \$limit provides genuine iteration limiting. Documented decision with the failing evidence preserved.

Restrictions enforced as the LRM demands (4.5.1): analog operators inside loops or solution-dependent conditionals are rejected with diagnostics that name the actual construct and cite the clause.

5. Analog behavior (LRM 5)

5.1 Analog blocks (LRM 5.2, 6.2) — ✓

Multiple analog and analog initial blocks per module execute as-if-concatenated in source order — verified including across instance flattening and with declarations interleaved between blocks.

5.2 Contribution statements (LRM 5.6) — ✓

Direct contributions with accumulation semantics, switch branches (V-then-I), indirect branch assignment (the LRM's ideal-op-amp construct), and port-flow probes:

```
V(out): V(inp, inn) == 0;           // indirect: drive out so V(inp,inn)=0
I(p,n) <+ V(p,n)/r;                // accumulates with other <+ statements
iport = I(<p>);                     // total flow through port p (ideal probe)
branch (<p>) pb;                    // named port branch: I(pb) == I(<p>) (E-84)
```

\$port_connected works on connected *and* unconnected ports of flattened instances (resolved per instance at elaboration time), and instantiating an undefined module is a hard error rather than a silent drop (both E-84). Bus actuals may be part-selects — positional, named, or width-1 onto a scalar port (E-85):

```
adc2 hi (out[3:2], in);             // positional slice
adc2 lo (.out(out[1:0]), .in(r));    // named slice
```

5.3 Procedural statements (LRM 5.7–5.9) — ✓

if/else; case over integers, reals, strings, and arrays (element-wise); all four loops with runtime trip counts that honor model-card parameter overrides; disable as the LRM's early-exit (Verilog-A has no break/continue, and the compiler says so with the disable idiom in the diagnostic):

```
for (i = 0; i < n; i = i + 1) acc = acc + w[i] * x[i];
repeat (4) y = f(y);
do begin y = y/2; end while (y > 1); // post-test: body runs once

begin : scan
  integer k;
  for (k = 0; k < 8; k = k + 1)
    if (v[k] > vth) disable scan; // the break idiom
end
```

5.4 casex / casez — ✓

The don't-care case variants, with the 2-state semantics the analog subset implies: x/z/? digits of a based literal written directly as an item form a comparison mask — the arm matches on every *care* bit. casex treats x/z/? as don't-cares, casez only z/?. The classic priority-encoder idiom (from the committed [examples/casexz_examples/](#)):

```
casex (req)
  4'b1xxx: grant = 8; // highest set bit wins
  4'b01xx: grant = 4;
  4'b001x: grant = 2;
  4'b0001: grant = 1;
  default: grant = 0;
endcase
```

Don't-care literals anywhere else, x digits under casez, and non-integer discriminants are located errors — no silent zeros.

5.5 Events (LRM 5.10) — ✓

@(initial_step) and @(final_step) genuinely gate (once at the start; once at the *successful* end, seeing the converged solution), with analysis-phase lists filtering by analysis type; cross/above/ timer with tolerances and persistent state; event OR lists:

```
@(initial_step("ac", "tran")) voff = 0.1;
@(cross(V(s) - vth, +1) or timer(0, 1u))
    n_events = n_events + 1;
@(final_step) $strobe("peak = %g at end of run", peak);
```

An operating point fires both step events (a single point is first and last); a failed analysis never fires final_step.

5.6 Analog functions (LRM 5.11) — ✓

Declared functions with input/output/inout arguments, arrays in all three roles plus array return values, locals (including array locals), loops inside bodies; inlined at the call site so derivatives flow through automatically:

```
analog function real dot;
    input real a[0:3], b[0:3];
    integer k;
    begin
        dot = 0;
        for (k = 0; k <= 3; k = k + 1)
            dot = dot + a[k]*b[k];
    end
endfunction
```

Recursion is illegal in Verilog-A; both direct and mutual recursion are clean errors naming the call cycle (mutual recursion formerly overflowed the compiler stack).

6. Hierarchy (LRM 6) — ✓

Module instantiation with named and positional ports and parameter overrides, open ports, instance arrays, arbitrary nesting, across `include boundaries; **generate** in all three forms with genvars usable in any expression; **defparam** with its LRM precedence over instance overrides; hierarchical names and \$root; implicit nets; net concatenation in port connections:

```
// genvars usable in any expression, e.g. a parameter override
module gpexpr(a, c);
    inout a, c; electrical a, c;
    genvar i;
    generate
        for (i = 0; i < 3; i = i + 1) begin : g
            res #(.r(1e3*(i+1))) ri (a, c);    // 1k || 2k || 3k
        end
    endgenerate
endmodule

defparam u1.r = 2e3;                // hierarchical override
defparam u1.u2.r = 4e3;            // two levels down; wins over #(...)
```


One boundary the OSDI target makes explicit (and the compiler explains in its diagnostic): structure binds at compile time — a generate condition or bound may depend on gensyms and literals but not on module *parameters*, because parameters arrive from model cards at simulation time, after structure is frozen. Loop trip counts and if arms, being behavior, may depend on parameters freely. [△](#) documented deviation with rationale.

7. System tasks and functions (LRM 9)

7.1 Display (LRM 9.4) — ✓

All five kinds (`$strobe`, `$display`, `$write`, `$monitor`, `$debug`) with the complete format surface — `[flags][width][.precision]` on every conversion, dynamic `*` widths, `%e/f/g/r` (engineering notation), `%d/h/o/b/c/s`, `%m`, `%%` — verified against exact `printf` output:

```
$strobe("Vth=%7.4f V   region=%2d   id=%r   name=%s", vth, reg, id, name);
```

7.2 File and string I/O (LRM 9.5, 9.8) — ✓

`$fopen`/`$fclose`/`$fdisplay`/`$fwrite`/`$fstrobe`/`$fmonitor`/`$fdebug`/`$fflush`/`$ftell`/`$fseek`/`$rewind`/`$feof` and `$swrite`/`$sformat`/`$sscanf`/`$fgets`/`$fscanf`:

```
integer fd;
analog @(final_step) begin
    fd = $fopen("iv.dat");
    $fdisplay(fd, "%g %g", vpeak, ipeak);
    $fclose(fd);
end
```

7.3 Random and distributions (LRM 9.13) — ✓ (deterministic semantics)

`$random`, `$arandom`, and every `$dist_*`/`$rdist_*` (uniform, normal, exponential, poisson, chi-square, t, erlang), verified against closed-form moments. [△](#) One deliberate semantic: draws are a deterministic function of (*seed*, *call site*) with no in-place seed advance — an advancing seed would return different values on every Newton iteration and destroy convergence. This gives reproducible Monte Carlo and independent per-instance variation, at the cost of in-evaluation sequences (which have no convergent meaning in an analog solver anyway).

```
parameter integer seed = 1;
real gain;
analog begin
    gain = 1.0 + sigma * $rdist_normal(seed, 0.0, 1.0); // per-instance mismatch
    ...
end
```

7.4 Simulation control (LRM 9.6–9.7) — ✓

`$finish`, `$stop`, `$fatal` are honored with simulator-correct timing (at the accepted-point boundary; `$finish` fires `@(final_step)` and ends the analysis cleanly; `$fatal` during setup reads as a configuration rejection), and `$warning`/`$error`/`$info` display.

7.5 The mechanism-unavailable group — [△](#)

`$test$plusargs/$value$plusargs` (ngspice has no `plusargs`), `$simprobe`, `$analog_node_alias`/`$analog_port_a` compile and return their LRM-specified fallback values (false / supplied default / 0). This is the LRM's own escape hatch for hosts without the mechanism; models using them run predictably instead of being rejected.

7.6 Attributes as simulator metadata — ✓

(* desc="..." *) on a variable exposes it as an operating-point variable (print @n1[gm], .save, .meas); (* type="instance" *) on a parameter puts it on the instance line and under alter/.dc sweeps; desc/units on parameters populate the OSDI descriptor.

8. Compiler directives (LRM 10) — ✓

The predefined source-location macros work (E-85): `__FILE__` expands to the source file's basename, `__LINE__` to the exact line (a use inside a ``define` body reports the definition site — documented textual-expansion semantics):

```
$strobe("evaluated at %s:%0d", `__FILE__, `__LINE__);
```

``define` with arguments (macros-in-macros, macro calls as arguments), ``ifdef`/`ifndef`/`elsif`/`else`/`endif` chained and nested, ``include` chains, ``undef``, ``resetall``, ``default_transition``, and the housekeeping set (``celldefine``, ``timescale``, ``line``, ``pragma``, ``default_nettype``, ...). Recursive macro expansion — direct or mutual — is a clean located error (formerly a compiler stack overflow), while the legal same-macro nesting inside *arguments* (``QUAD(x) = `TWICE(`TWICE(x))`) expands finitely:

```
`define TWICE(x) ((x)*2.0)
`define QUAD(x) (`TWICE(`TWICE(x))) // same-macro nesting in ARGUMENTS: legal
`ifdef HIGH_PRECISION
    `define TOL 1e-12
`else
    `define TOL 1e-9
`endif
```

(*`default_discipline* is AMS-only and out of scope — implicit nets themselves are Verilog-A and work, taking their discipline from the connected port.)

9. Compliance summary

LRM area	Status	Verified by
2 Lexical (identifiers, numbers, strings, attributes)	✓	escid, stresc, comment suites
3.2–3.3 Value types, variables, persistence	✓	vartype, variable_persistence, intstate, varinit
3.4 Parameters (ranges, localparam, aliasparam, arrays, paramset)	✓ (default-exemption ⚠ documented)	paramrange, localparam, array, paramset, paramsethsp
3.5–3.7 Natures, disciplines, nets, buses, nodesets	✓	derivednature, domainbind, bus, netinit, ground, signalflow
4.1–4.3 Operators, precedence, functions	✓ (audited)	operator, precedence, shift, concat
4.5 Analog operators (all)	✓ (limexp stateless ⚠ documented)	absdelay, laplace, zi, slew, transition, idt*, ddx, opargs, discontinuity, last_crossing
4.6 Noise (incl. correlation 4.6.4)	✓	noise, noisecorr, noisejw
5.2/6.2 Analog blocks (multiple)	✓	multianalog
5.6 Contributions, indirect assignment, port flow (incl. named port branches)	✓	indirect_assignment, portflow, signalflow, lrm

LRM area	Status	Verified by
5.7–5.9 Procedural statements, loops, disable	✓	analogloop, dowhile, repeat, disable, arraycase
casex/casez	✓	casexz
5.10 Events (steps, cross/above/timer, OR lists, phase lists)	✓	initial_step, finalstep, cross, timer, lrmcorner
5.11 Analog functions (arrays in/out/return)	✓	funcarray, arrayout, arrayret
6 Hierarchy (instantiation, generate, defparam, \$root, part-select connections)	✓ (param-shaped structure ⚠ explained)	instantiation, generate, defparam, hiername, implicitnet, partselect
9.4–9.8 Display, file/string I/O	✓	display, fileio, stringio
9.13 Random/distributions	✓ (deterministic seed ⚠ documented)	rng, montecarlo
9 misc (\$finish family, \$simparam, attributes)	✓	simctrl, simparamstr, opvar
plusargs / simprobe / node aliases	⚠ LRM fallbacks	alias
10 Compiler directives (incl. `__FILE__`/ `__LINE__`)	✓	preproc, directive, defaulttransition, filemacro
Mixed-signal / digital (outside Annex C)	× by scope	—

As of E-84 the standard's own examples are a compliance suite: every code example in the LRM 2023 PDF is extracted and compiled (examples/lrm_examples/ — 39 compile, 20 documented limitations pinned to their diagnostics, 21 correctly rejected as mixed-signal), and the 146 non-module fragments double as a no-crash fuzz corpus.

Beyond construct-level checks, three cross-cutting guards pin that the *whole language implementation* produces correct physics: the 92-model industry corpus compiles and runs, the static/dynamic/thermal physics suites recover textbook device laws through the full pipeline (60 mV/dec, $C \equiv dQ/dV$ across analyses, $E_g \approx 1.25$ eV from a compiled MEXTRAM), and the twin benchmarks show OSDI output matching hand-coded simulator devices to machine precision.